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Polaron formation via doping process in an organic semiconductor polymer based on thiophene-phenylene

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ABSTRACT

In this research, we synthesized and analyzed the polymer 5-(5-(10-chlorodecyloxy)-2-methoxy-4-(5-methylthiophen-2-yl)phenyl)-5′-(2-(2-ethylhexyloxy)-5-methoxy-4-(5-methylthiophen-2-yl)phenyl)-2,2′-bithiophene (PTBT46). The polymer was dissolved in chloroform and subjected to gamma radiation doses ranging from 0 to 100 Gy (gray). We observed that PTBT46 can be doped with H+ions, as indicated by visible color changes, optical measurements, and a tenfold increase in conductivity. When analyzing PTBT46 in solution using THF (tetrahydrofuran) and chloroform, we found a notable solvent dependence. In chloroform, new absorption bands appeared at approximately 680 nm and 2630 nm after radiation doses of 25 to 100 Gy, which were not present in THF. These bands are attributed to polarons and bipolarons resulting from the doping process. Additionally, we observed a spectral "blue shift" with increasing H+concentration due to changes in molecular conjugation, without degradation of the polymer chain. This shift, confirmed by Fourier-transform infrared (FTIR) spectroscopy, is reversible and suggests potential applications in optoelectronic devices and advanced radiation sensors.

1. Introduction

Synthetic polymers have revolutionized the 20th century, particularly with the discovery of organic semiconducting polymers in 1970 by Heeger et al. through chemical doping of polyacetylene [1]. Since then, numerous polymers have been synthesized to enhance or discover novel physicochemical properties that can enhance the performance of devices based on organic compounds. Among them, polymers based on the thiophene-phenylene-thiophene (PTBT) aromatic groups have emerged, combining the thiophene and phenylene electronic properties in the same chain with the inherent mechanical properties of polymers [2].

The polymeric electronic properties can also be altered by adding or substituting side groups in the main chain, achieved through controlled chemical synthesis. The primary method of doping organic semiconductors involves oxidation (p-doping) or reduction (n-doping) of the conjugated chain. Although the mechanisms of electronic transport are not fully elucidated, especially for monomers containing aromatic groups, their use is acknowledged to enhance conductivity [1,3–6]. Despite the conductivity of organic materials being less effective than inorganic semiconductors, the advantages of low processing costs, flexibility, and ease of fabrication make them appealing from technological and commercial standpoints.[7] Techniques such as chemical and electrochemical doping of conjugated polymers have been extensively employed to significantly improve the conductivity of these materials by several orders of magnitude [8,9]. However, chemical doping is challenging to control, and electrochemical doping results in small amounts of effectively doped material, requiring complex purification methods [10,11]. Therefore, the control, development of new

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techniques or processes, and understanding of the doping mechanisms in organic semiconductors become a fundamental issue with potential large-scale commercial applications [12–14].

The polymeric interaction with gamma radiation has been studied since the 1970 s, and different effects have been observed depending on the polymer's chemical structure and the energy range used for the irradiation process. Reports have indicated various polymer degradation effects, such as cross-linking [15], scission and photodegradation [16]. While the mechanisms involved in the interaction of gamma radiation with polymers are not entirely elucidated, changes in conductivity and optical properties have been particularly observed in polyaniline [17], poly-p-phenylene vinylene (PPV) and its derivatives [18–20], and polyhexylthiophene (P3HT) [21]. The main result obtained with irradiated P3HT devices was a significant improvement in conductivity with increasing gamma radiation dose. In general, in polythiophenes, it is possible to create polaronic, bipolaronic, or neutral states within the main chain [22,23]. Similar to studies on MEH-PPV, where the outcome was limited to the use of the polymer in a solution due to the effect being solvent-dependent. In other words, the effect is indirect, as radiation breaks the solvent chain, and the radicals resulting from solvent chain break attack the polymer chain. The attack occurs in the vinylene, breaking the double bond and consequently leading to the disruption of conjugation, as observed in the blue shift in the optical absorption bands (AO). This experimental result has already been supported by theoretical studies, and the mechanism of attack in the vinylene is well-established [24].

However, the use of vinylene-based polymers is not ideal due to their degradation of the main chain. In search of applications for the studied materials and observing that polymeric materials, when dissolved in halogenated solvents, can be sensitive to ionizing radiation, mainly because these solvents have a significant cross-section for interaction with ionizing radiation..[19,20,24,25] Recent results using low doses of gamma radiation (<100 Gy) in a halogenated solution of poly[2-methoxy-5-(2′-ethyl-hexyloxy)-p-phenylenevinylene] demonstrate doping effects. [26,27] Raval et al. reported the pentacene doping via gamma radiation using lower radiation doses (20–100 Gy). [28] However, pentacene faces significant solubility issues, which limit its use due to higher costs compared to conventional film deposition processes.

Luminescent polymers containing TB (thiophene-phenylene) groups within the same chain were initially proposed by Heeger and colleagues to address the low quantum efficiency issues observed in thiophene-based compounds. [29] The incorporation of phenylene-type compounds into the main chain enhances optical properties while preserving the electrical characteristics derived from thiophene.[29,30] TB compounds exhibit high emission efficiency [29] and good solubility with the introduction of alkyl side chains,[31,32] variations in optical properties due to conformational changes induced by the insertion of alkoxy side chains, [29,33] as well as excellent adhesion properties and good film formation. [2].

In this context, our research explores the doping effect on the conjugated polymer poly[5-(5-(10-chlorodecyloxy)-2-methoxy-4-(5-methylthiophen-2-yl)phenyl)-5′-(2-(2-ethylhexylo-xy)-5-methoxy-4-(5-methylthiophen-2-yl)phenyl)-2,2′bithiophene] (PTBT46). Previous studies [21,22,24,25] have discussed that the presence of H+ions in halogenated solvents, caused by the exposure of the sample to gamma radiation, leads to the formation of polaronic and bipolaronic bands, as well as chromatic alterations. This effect is associated with the reduction in conjugation length promoted by the attack of the radicals generated during the radiolysis of the solvent on the side chains, in S…O bonding regions. Given the complexity of the processes involved in this mechanism, the influence of the solvent on the doping process was investigated. Notably, while aromatic groups are known for their resilience to radiation,[34] the role of the side chains becomes pivotal in their interaction with gamma radiation, primarily driven by steric effects. [35] Our findings revealed a remarkable transformation in coloration, pH sensitivity, as well as alterations in its optical and electrical properties when PTBT46 was solubilized in halogenated solvents. However, it is essential to recognize that ionizing radiation does exert an influence on the thiophene-based aromatic groups, as seen in the case of P3HT. Furthermore, the side chains also contribute significantly to the interaction dynamics with gamma radiation. The study was developed by conducting its photophysical characterization using a range of techniques, including UV–Vis absorption, photoluminescence (PL), Fourier-transform infrared spectroscopy (FTIR) and Raman spectroscopy.

2. Materials and methods

2.1. Methodology

Optical Characterization: A Shimadzu UV-3600 UV–Vis-NIR (185–3300 nm) spectrophotometer were employed for Optical Absorption (AO) and in-situ, measurements. The in-situ experiment was performed coupled with an Ivium CompactStat potentiostat in chronoamperometry mode together with an electrochemical cell (for experimental characteristic see SI – Supplementary Information).

Structural Characterization: Immediately following the material synthesis, Fourier Transform Infrared Spectroscopy (FTIR) measurements were conducted using a Perkin Elmer FTIR 1000 spectrometer, spanning the range from 400 to 4000 cm^{-1}, in attenuated total reflectance (ATR) mode. ^1H NMR spectra were recorded in $CDCl_3$ at 400 MHz using a Bruker AC 400 spectrometer. Raman spectra was recorded on a multichannel Jobin-Yvon T64000 spectrometer connected to a cooled CCD detector. To analyse the irradiation process, FTIR measurements were performed using a Nexus 470 FT-IR instrument, covering the same spectral range. Films were obtained by drop-casting irradiated solutions onto NaCl substrates, forming a film after solvent evaporation.

Electrical and Electrochemical Characterization: The investigation of the electrical behaviour (DC electrical measurements) to observe changes in material conductivity with varying doses of gamma radiation was conducted using a commercial interdigitated electrode (IDE) (see Figure SI1). A Keithley 2400 voltage source was employed to acquire current–voltage (I vs. V) measurements (more detail see Supplementary Information (SI)).

Irradiation Process: The irradiation procedures were carried out in collaboration with IPEN (Instituto de Pesquisas Energéticas e Nucleares)/USP at the Laboratory of Intense Radiation Sources (LFIR) within the Center for Radiation Technology (CTR).

2.2. Materials

The PTBT46 polymer was synthesized through a Stille cross-coupling reaction catalysed by $Pd(PPh_3)_4$ (Palladium-tetrakis(triphenylphosphine)) (Sigma-Aldrich CAS Number 14221–01-3) between 5,5′-bis (tributylstannyl)-2,2′-bithiophene (compound 2 in Fig. 1 – Sigma-Aldrich CAS Number 171290–94-1) and two dissymmetric dibrominated dialkoxy benzenes: 1,4-dibromo-2-(11-chloroundecyloxy)-5-methoxybenzene and 1,4-dibromo-2-(2-ethylhexyloxy)-5-methoxybenzene (compounds 1 and 3 in Fig. 1, respectively – Sigma-Aldrich CAS Number 209625–37-6). The synthesis of these dissymmetric monomers was achieved through an etherification reaction with the corresponding alcohol, using a Mitsonobu reaction on 4-methoxyphenol. Subsequent dibromination at the 2 and 5 positions yielded the respective starting dibrominated monomers for compounds 1 and 3, with moderate and preparative yields, respectively. The Stille reaction was conducted under an inert atmosphere of argon, using water-free solvents, and $Pd(PPh_3)_4$ as the catalyst (0.015 eq.). One equivalent (1 eq) represents the exact stoichiometric amount of a reagent needed for the reaction, based on the balanced chemical equation. To create the regular alternated copolymer PTBT46, one equivalent (1 eq) of compound 2 was combined with half an equivalent (0.5 eq) of each of the two dibrominated monomers, compounds 1 and 3. Monomer 1 is characterized by a single methoxy

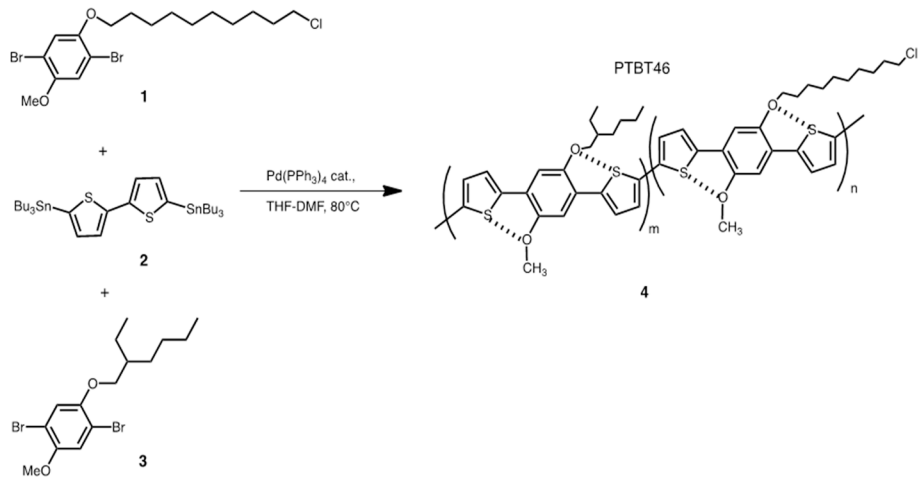

Fig. 1. Chemical rote of PTBT46.

group and a decyl lateral chain terminated by a chlorine atom, designed to enhance the gamma radiation cross-section.

The compound called monomer 3 is also substituted with a methoxy moiety and a 2-ethyl-hexyl branched chain in order to improve the solubility of the resulting polymer. After 72 h of reaction in THF/DMF (50:50) (tetrahydrofuran/dimethylformamide PA/ACS 100 % – Synth) at 80 °C, under reflux conditions and protected from light, the obtained polymer was purified by precipitation in methanol (99.5 % PA) and left to rest for 48 h away from light. The solvent was removed by centrifugation, and the product was dried under vacuum.

After the synthetic step, the PTBT46 was subjected to characterization using standard analytical techniques. Further details regarding the synthesis and elemental measurements can be found in sections 2 and 3 of the Supplementary Information. For absorption measurements subjected to gamma radiation and electrical measurements, a more concentrated solution (10^{-3} M) was used due to the possibility of degradation processes. For the remaining characterization measurements, a more diluted solution (10^{-5} M) was used to avoid saturation in spectroscopic measurements, in addition, decreasing the probability of molecular aggregation and it contains more free sites. The ^{1}H NMR spectrum (Figure SI2), recorded at 400 MHz, and the infrared spectroscopy data for the four compounds are consistent with the polymer structure depicted in Fig. 1. The Raman measurement was performed in the back scattering configuration using an Ar-ion laser (Spectra Physics 2000) with 632.8 nm exciting radiation at room temperature (Figure SI3.a). Optical measurements were conducted using a $CHCl_3$ solution (10^{-5}M). The absorption spectrum revealed a π-π * transition with a maximum at 468 nm (insert Fig. 4).

Table 1 presents a general summary of the data obtained from optical characterization in solution, cyclic voltammetry (CV) measurements, and gel permeation chromatography (GPC). From the analysis of the absorption spectrum, the optical gap was estimated to be approximately 2.6 eV (Figure SI4.a). The oxidation (HOMO) and reduction (LUMO) potentials of the polymer were obtained from the CV curves, with values of −4.55 eV and −2.16 eV, respectively (Figure SI5). SEC analysis indicated the presence of a relatively low-dispersity polymer (PDI=1.39), consisting of six repeating units, corresponding to 104 aromatic cycles, with a 23 conjugation degree, a molar mass of 20,849, and with an average molar mass number of 14,931. The fluorescence quantum yield Φ_F, obtained using photoluminescence (PL) (Figure SI4.b), was 0.76 (76 %) as evaluated through a dilute solution in THF.The standard sample used was fluorescein solubilized in ethanol, which emits light in the same range as the copolymers, from 500 to 600 nm.

3. Results and discussion

The PTBT46 solutions in chloroform ($CHCl_3$) and in tetrahydrofuran (THF), both with concentration at 10^{-3} M were irradiated with doses of 0 Gy, 25 Gy, 50 Gy, and 100 Gy of gamma radiation at a rate of 1.55 KGy/h (Fig. 2). A visual color change is observed, shifting from orange to green in the solutions of PTBT46 diluted in chloroform, while this phenomenon is not observed in those diluted in THF. This change in colour intensifies with increasing gamma radiation dose. The change in colour of conjugated polymers from red to blue is a well-known phenomenon in the literature, attributed to the interaction between gamma radiation and halogenated solutions of conjugated polymers [16,24,24,36]causing material degradation. However, the colour shift from red to green observed in PTBT46/$CHCl_3$ indicates that the interaction occurs in a completely different manner from what has been observed until now. Comparing the absorbance behaviour of the two samples presented in Fig. 2, it is noticeable that no change in line shape with respect to the received gamma radiation dose was observed for the PTBT46/THF sample. However, for the samples dissolved in chloroform (Fig. 2a), we observe the emergence of two new absorbance bands centered at 680 nm (1.82 eV) and 2630 nm (0.47 eV) when subjected to gamma radiation. The peaks marked with (*) in the figure details are related to the optical absorption of the solvents and the quartz cuvette (for details, see Figure SI6 and S17 in the supplementary information). The reason for any change in the absorption line shape spectrum with respect to the received gamma radiation dose for the PTBT46/THF sample is a consequence of the solvent effect. The gamma radiation interaction with polymeric solutions in halogenated solvents, such as chloroform, is different compared to polymer solutions in other types of solvents, like THF, which is a polar aprotic solvent. The appearance of polarons and bipolarons results from chemical attack on the polymer's main chain by radicals present in the solution. In the system studied here, the radicals are generated from the breakdown of the halogenated solvent chain by gamma radiation, i.e., Cl- and CHCl2• from chloroform,

Table 1
Basic characterization data of PTBT46.

Absorption peak	Gap*	HOMO	LUMO	M_n	M_w	PDI	DP	Φ_F
468 nm	2.617 eV	−4.55 eV	−2.16 eV	14.931	20.849	1.39	23	0.76

* Estimated through optical absorption spectra. (Figure SI14.a).

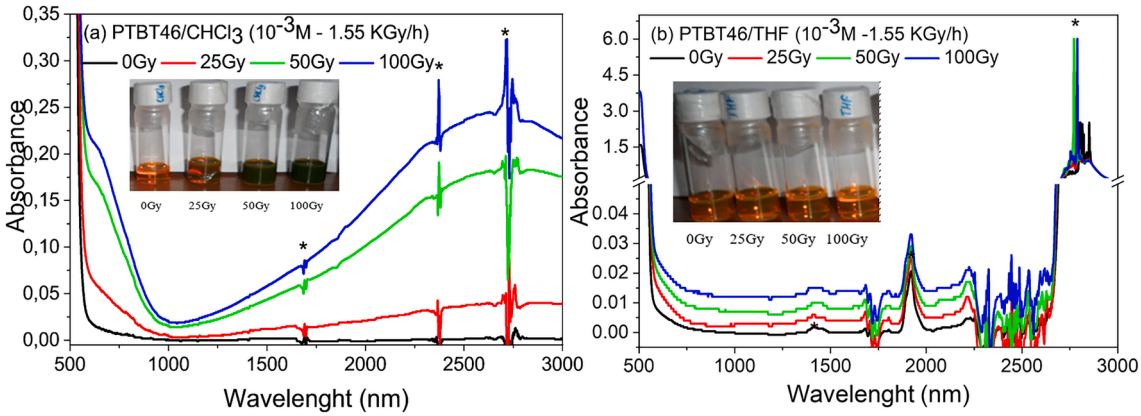

Fig. 2. Optical absorption spectra UV–Vis-NIR of PTBT46 in chloroform (a) and in THF (b), both in function of dose radiation: 0 Gy, 25 Gy, 50 Gy and 100 Gy, with the intensity ratio 1.55 KGy/h. The inset figure displays the visual effect of the radiation dose on both solutions (in THF, the spectra were shifted for better visualization).

and the subsequent recombination and formation of HCl, as will be shown later. In the THF solvent, there is no interaction with gamma radiation and consequently no formation of radicals to attack the polymer chain, thus preventing the doping process.

In order to characterize the doping process of PTBT46, in situ UV–Vis-NIR measurements were conducted on PTBT46 films obtained from the polymer solution in $CHCl_3$ (10^{-5}M). The results are depicted in the spectra in Fig. 3. The band around 0.8 eV exhibited a higher signal-to-noise ratio, likely due to the PTBT46 films being deposited onto FTO, which was found to introduce noise in the near-IR measurement. However, this band remains analysable without significant hindrance, as shown in Fig. 3.

Under the cathodic potentials (negative potential) application, no significant changes were observed in the spectral bands ranging from 340 nm to −1982 nm (0.625——3.625 eV). It's important to note that these spectra overlap with those at 0 V and + 0.2 V. However, during the application of anodic potentials (positive potential), significant modifications were observed in the main polymer bands above 2.2 eV, accompanied by the emergence of bands below 2.2 eV corresponding to polarons and bipolarons. The transition related to the thiophene-thiophene aromatic rings (thth)[37] is associated with the first band (<350 nm or 3.5 eV). Although it cannot be fully observed, a notable increase in its line shape and, consequently, its intensity is evident with

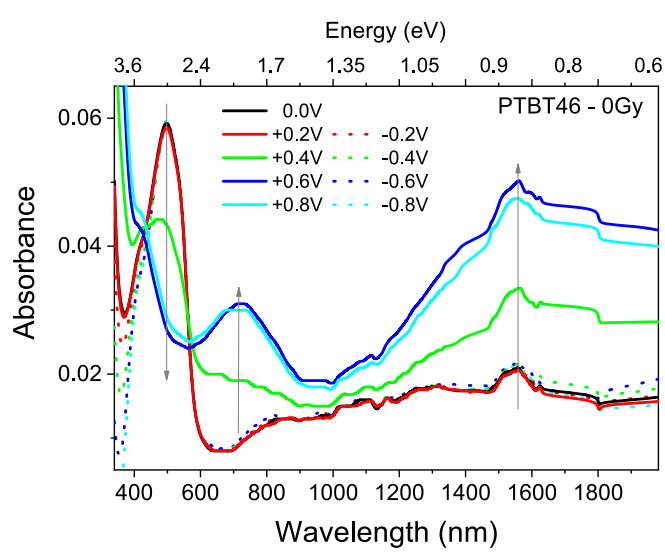

Fig. 3. In situ UV–Vis-NIR spectra of the FTO/PTBT46 film obtained from the polymer solution in $CHCl_3$ (10^{-5}M).

increasing potential (resulting from the insertion of perchlorate anions (ClO_4^-) into the polymer film). The band centered at 410 nm (3.02 eV), related to conjugation and the electron-donating effect of the alkoxy group in the monomer [38,39] remains relatively stable with varying applied potentials.

However, the band around 495 nm (2.51 eV), associated with the thiophene-phenylene (thph) transition[37] is highly sensitive to the doping process. It intensity decreases as the polymer chain undergoes oxidation, simultaneously with the emergence and intensification of bands attributed to polarons and bipolarons (bands < 2.2 eV). There is an isosbestic point at (~560 nm) where these bands effectively appear, starting from an applied potential of 0.4 V, with a significant jump from 0.6 V to 0.8 V. This behavior is attributed to the interaction of the anion (ClO_4^-) with the methoxy group, particularly close to the S-O interaction. This interaction interferes with and oxidizes the material. The oxidation primarily occurs in the thiophene unit, leading to chain distortion and the formation of polaronic and bipolaronic bands. Studies related to the doping of polythiophenes [22,23] assert that after undergoing doping processes, polythiophenes transition into an oxidized state, thus forming polarons. Subsequently, they stabilize into a bipolaronic and neutral state within the chain, remaining in the oxidized state. [26].

Studies involving polythiophenes have demonstrated the effect of doping through electrochemical processes, leading to the formation of polaronic and bipolaronic bands. [22,23] This behavior is analogous to the process observed in the thiophene and phenylene-based derivative used in this work. Through the analysis of the measurements conducted, it can be inferred that the outcome of gamma radiation interaction with the polymer, as observed in the UV–Vis-NIR in situ measurements, follows the same pattern as the emergence of the two bands in both spectra. Consequently, this is indeed a polymer oxidation process induced by radicals produced by gamma radiation in the halogenated solvent, resulting in doping within the thiophene moiety of the polymer's main chain, as observed in previous works..[22,23].

Doping processes of polymeric chains with chemical structures similar to that in this work, carried out through electrochemistry, have also shown a band located around 680 nm, attributed to bipolaronic transitions.[2] The formation of polaron and bipolaron states results from chemical attack (oxidation or reduction) on the polymer's main chain by radicals present in the solution. Since no significant changes were observed in the absorption spectra in THF or chloroform within the $\pi \rightarrow \pi^*$ transitions region (<600 nm in Fig. 2), the only possibility is that radicals generated by gamma radiation in chloroform are interacting with the polymer chain. This is possible due to the breaking of the halogenated solvent molecule by gamma radiation, in the case of chloroform forming Cl^- and $CHCl_2\bullet$ (equation (1), followed by subsequent recombination with H^+ to yield HCl (equation (2).[40]

$$CHCl_3 + \gamma \rightarrow Cl\bullet + CHCl_2 \quad (1)$$

$$Cl\bullet + CHCl_3 \rightarrow HCl + CCl_3 \quad (2)$$

The interaction of chloroform with ionizing radiation causes the formation of HCl, according to equation (2), whose ionizable hydrogen can be easily detected by potentiometric sensors, with a pH meter being the most suitable for confirming this change in ionic potentials. The pH meter operates by comparing the potential of an electrode, where charge accumulation will occur, with the reference electrode (Ag/AgCl in 3 mol/L KCl at a fixed potential of 0.198 mV relative to the hydrogen electrode at 25 °C). However, the pH scale is designed for use in aqueous media and may not be applicable in solvent mixtures or organic solvents, necessitating the establishment of a new scale for this solvent in relation to hydrogen potential.[41] Thus, any ionic variation in the solution, even in a non-aqueous or organic solvent, will be registered as a potential change by the pH meter. Therefore, the increase in the dosage of incident radiation in the solution leads to an increase in HCl originating from chloroform. The pH values obtained were 6.5 for the non-irradiated solution and completely acidic (pH=1.6) for the 100 Gy dose, as shown in Table 2.

As shown in equations (1) and (2), the gamma radiation-induced breakdown of chloroform results in Cl• and $CHCl_2$• radicals in the first step, followed by the formation of HCl in the second step. These radicals resulting from chloroform breakdown are the likely dopants for the polymer chain. To confirm this hypothesis, solutions of PTBT46 (10^{-5} M) in THF were gradually added to increasing amounts of HCl, and optical absorption measurements were performed. The absorbance spectrum of PTBT46/THF does not exhibit polaron or bipolaron bands, as discussed earlier and shown in Fig. 2b. However, as the concentration of HCl in the solution increases, the spectrum behaves similarly to what was observed in the interaction of PTBT46/$CHCl_3$ with gamma radiation, as shown in Fig. 4. Due to the strong water absorption above 900 nm in HCl, the analysis and comparison were restricted to the range from 320 nm to 900 nm.

Fig. 4 illustrates the decrease in the main band (~470 nm) and the simultaneous increase in the band at 680 nm, exactly as observed for PTBT46/$CHCl_3$ under the influence of gamma radiation (Fig. 2a and insert of Fig. 4). Furthermore, the blue shift observed in the absorption spectra of the irradiated polymer is also slightly noticeable in the HCl treatment. This difference can be explained by the fact that the amount of HCl may not have been sufficient to induce the effect to the same extent as high-energy radiation. High-energy gamma radiation induces solvent molecule breakdown, but as the radiation dose increases, gradual blue shifts are observed in the spectra in Figs. 2 and 4, indicating polymer conjugation disruption, likely due to chain torsion. Note that the intensity around 495 nm (2.51 eV), which corresponds to the band with an ill-defined line shape in the spectrum, is related to the thienophenylene (thph) transition [37] and is highly sensitive to the doping process. The main band shifts towards the blue due to the decrease in the contribution at 495 nm, a consequence of torsion/breaking of the S…O interaction along with effective conjugation disruption. This is also supported by the decrease in its absorption intensity as the polymer chain undergoes oxidation, while the bands attributed to polarons and bipolarons emerge and increase in intensity, as observed in Fig. 4 and

Table 2
Ph values of ptbt46/chloroform solutions in function of radiation dose: 0 Gy, 25 Gy, 50 Gy and 100 Gy with the ratio of 1.55 KGy/h.

PTBT46/$CHCl_3$	
DOSE (Gy)	pH
0	6.5
25	5.9
50	5.1
100	1.6

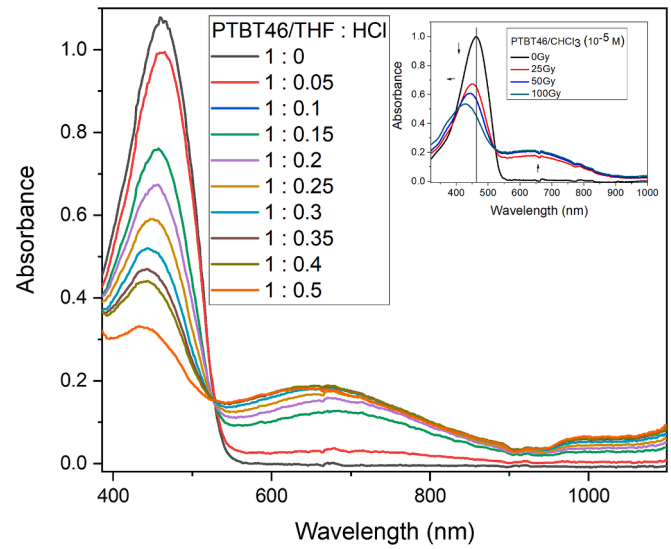

Fig. 4. Optical absorption spectra as a function of the addition of HCl to the 10^{-5}M solution of PTBT46/THF (volume ratio). The inset figure displays the optical absorption spectra of the 10^{-5}M solution of PTBT46/$CHCl_3$ before and after the irradiation process.

Figure SI8.

It is important to note the presence of isosbestic points in Fig. 3 (~430 nm and ~ 560 nm), in Fig. 4 (~530 nm), and in the inset of Fig. 4 (~400 nm and ~ 530 nm), where the absorption of different species is the same at these wavelengths. The presence of these two isosbestic points is evidence of the H+attack on the polymer chain, and consequently, the formation of polarons and bipolarons in the doping process. The first point refers to the break in the effective conjugation of the polymer's main chain with the torsion of the S…O interaction, while the second point is related to the oxidation process of the polymer chain during the system's doping process. With the pH varying from 6.5 to 1.6 as the radiation dose increases, the same behavior was observed in the experiment of adding HCl to the polymer solution and also in the in situ absorption experiment, corroborating the claim of polaron and bipolaron formation in the chain. These points demonstrate the presence of two or more absorbing species, showing the equilibrium between parts of the molecule that underwent the H+ion attack on the main chain and others that have not yet undergone this process.

The halogenated solvent interaction process on polymer chain after gamma irradiation has been previously reported by Graeff and colleagues for MEH-PPV, in which radicals preferentially attack the double bond of the vinylene group, leading to the disruption of the polymer's conjugation and shifting the absorption spectra of the $\pi \rightarrow \pi^*$ band toward the blue.[24,25] Theoretical studies confirm that unsaturated regions in the main chain are more susceptible to radical attack, resulting in chain twisting rather than chain breaking, especially in compounds with a fully aromatic main chain, such as PTBT46, as opposed to compounds like MEH-PPV.[42] Theoretical studies and NMR measurements indicate that other sites in MEH-PPV are preferentially targeted by radicals from halogenated solvents,[19,20,25] as shown in Fig. 5a. However, the incorporation of chlorine at these sites does not induce significant changes in the absorption spectrum of MEH-PPV. Nevertheless, as PTBT46 and MEH-PPV share a similar chemical structure with phenylene groups and alkoxy side chains, it can be assumed by analogy that these sites in PTBT46 are also susceptible to radical attack. A comparison of the chemical structures of MEH-PPV and PTBT46 is shown in Fig. 5.

The significant difference with PTBT46 lies in the absence of vinyl groups and the presence of thiophene groups in the main chain, leading to different types of interactions between radicals and the polymer. The

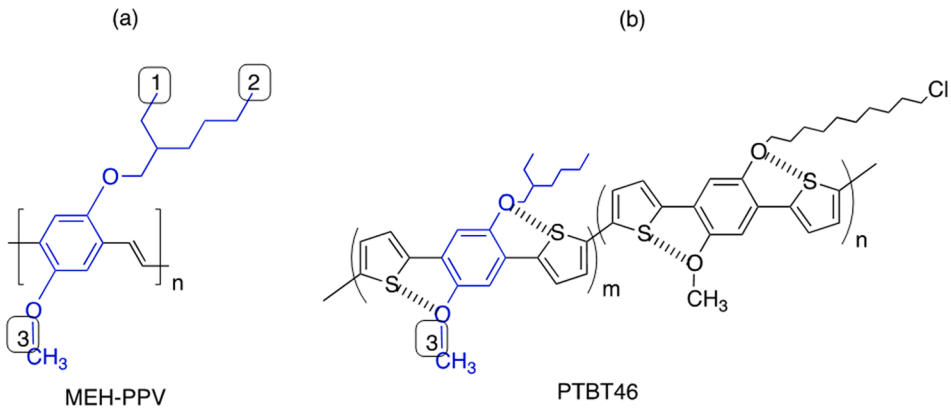

Fig. 5. Chemical structure of polymers with similar regions highlighted in blue: (a) MEH-PPV monomer with regions of side branches preferred for chemical attack by halogen radicals and (b) chemical structure of PTBT46.

main process observed was likely the doping of the chain, which probably occurred after the radicals attacked near position 3 of PTBT46 (Fig. 5), causing a disruption of the S…O interaction and resulting in chain twists due to the loss of stability provided by this interaction.

Based on this results, it was possible to propose the energy diagram presented in Fig. 6, where the observed phenomena are comparable to the doping effects observed in conjugated polymers such as polyaniline and P3HT, involving the formation of polarons and bipolarons. [2,43,44] In Fig. 6, we illustrate the main electronic transitions: i) π-π* transition of the neutral polymer (~2.47 eV), ii) polaronic transition (~1.75 eV), and iii) bipolaronic transition (~0.76 eV).

In order to verify if there is a conjugation break or conformational change in PTBT46 (blue shift – Figs. 2 and 4) due to the observed blue shift in the optical absorption measurements, FTIR measurements were conducted, as shown in Fig. 7. The FTIR spectra of PTBT46 irradiated at a dose rate of 1.55 KGy/h and doses of 0 Gy, 25 Gy, 50 Gy, and 100 Gy are offset in the figure for better visualization.

The band intensity related to the vibrational mode at 850 cm^{-1}, attributed to the in-plane vibration of the C-S-C in the thiophene ring, was fixed to assess changes in the line shape of the other bands due to the increase in protons (H+). It was observed that even for the highest dose (100 Gy), all bands were maintained, with no appearance of any new bands. However, some infrared absorption bands showed broadening or changes in relative intensity, such as the bands related to the symmetric stretching mode (1214 cm^{-1}) and asymmetric stretching mode (1272 cm^{-1}) of the C–C between rings (Fig. 7b) and the bands related to the asymmetric stretching mode (1380 cm^{-1}) of the C-H_3 and asymmetric stretching mode (1461 cm^{-1}) of the R-O-CH_3 (Fig. 7c). [38]These modifications support the model proposed in this study that the attack of radicals resulting from the radiolysis of chloroform (Eq. (1) and (2) occurs in the methyl group. As this methyl group is linked to the oxygen responsible for the S…O interaction, the doping of PTBT46 should be breaking this interaction, which explains the changes in relative intensity of the asymmetric stretching of C–C between rings at 1272 cm^{-1} and, consequently, the decrease in the effective conjugation of the polymer observed through the blue shift of the absorbance bands (Figs. 2 and 4).

This is a directly related change to conformational defects and not to the polymer degradation chain, as observed for MEH-PPV. The analysis of the 50 Gy dose also shows changes in other vibrational modes at 757, 786, 875, and 1737 cm^{-1} related to the stretching of C-Cl, C-H_{Th}, C-O-C, and C-O, respectively. Modifications in stretches related to the benzene ring (1033, 1594, and 2959 cm^{-1}) are also noticeable and dose-dependent. The analysis about FTIR measurements, corroborates the hypothesis of PTBT46 effective conjugation alteration due to conformational changes. For a more comprehensive and precise analysis of the molecular structure, confirming the effective polymeric conjugation decrease and demonstrating that there is no degradation of the polymer chain, see the NMR measurements in the Supplementary Information (Figure SI9.a). These changes in molecular conjugation and the appearance of polaronic bands are a reversible process (see Figure SI9.b). Another important result is that, in addition to the conformational modification of the polymer chain, there is no chain breakage. In other words, gamma radiation promotes the modification of the main chain to a quinoid type of PTBT46. These quinoid structures are well-known in polymers where polaron and bipolaron formation occurs. [45,46] Fig. 8a shows the chemical structure of non-irradiated PTBT46, and Fig. 8b shows an idealization of the quinoid structure of PTBT46.

It is well known that material chemical doping can increase its conductivity, as observed in the case of P3AT.[47] Therefore, it is

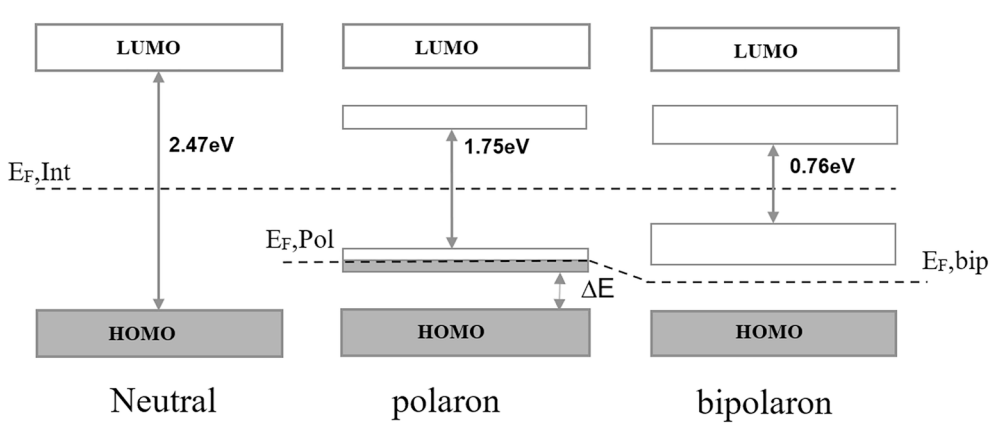

Fig. 6. Proposal band structure of polaronic PTBT46 in solution.

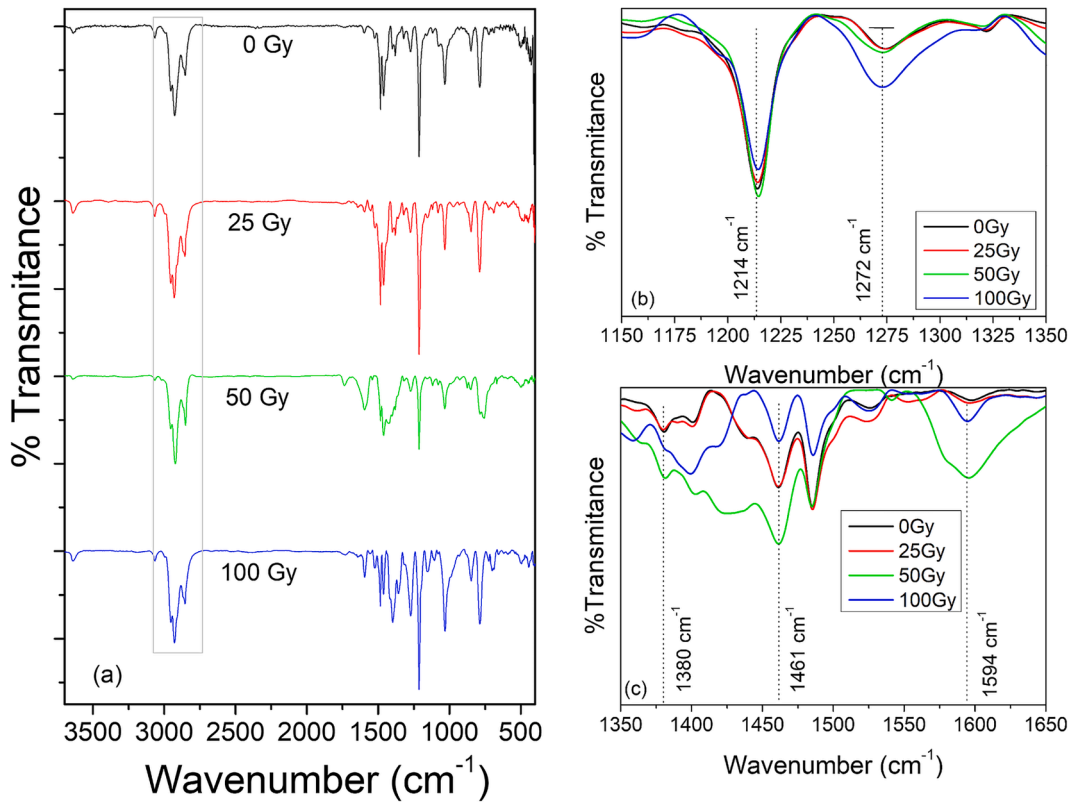

Fig. 7. (a) FTIR spectra normalized to unity at 2923 cm^{-1} and vertically shifted for better visualization of PTBT46/$CHCl_3$ samples irradiated at doses of 0, 25, 50, and 100 Gy. FTIR spectra in the scales: (b) from 1150 to 1350 cm^{-1} and (c) from 1300 to 1550 cm^{-1}.

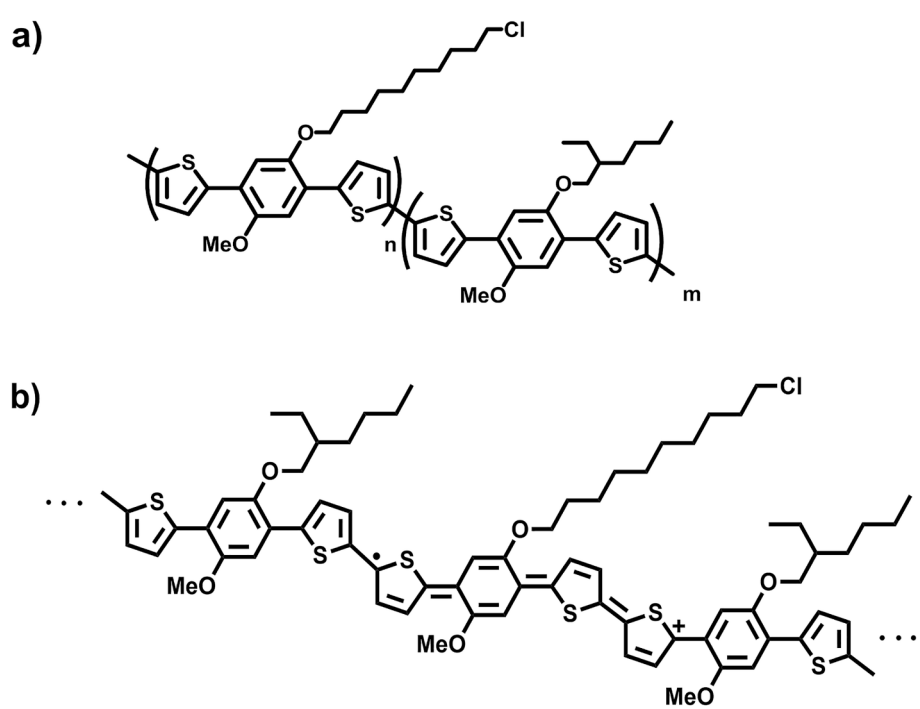

Fig. 8. PTBT46 structure: a) original and b) quinoid (idealized).

expected that the PTBT46 doping presented in the previous analysis will lead to an increase in the material's conductivity. Thus, the previously outcomes relies on the needed to conduct study of the PTBT46 electrical conductivity as a function of gamma radiation dose was conducted. For this purpose, a PTBT46 film was deposited via spin coating onto a platinum IDE. The films were prepared from solutions irradiated with doses of 0 Gy and 100 Gy at a concentration of 10^{-3}M, and their I-V curves are presented in Figure SI10. The electrical conductivity of the film irradiated with a dose of 100 Gy ($\sigma = 5.65 \times 10^{-3}$ S/m) showed an increase of one order of magnitude compared to the non-irradiated

sample ($\sigma = 2.68 \times 10^{-4}$ S/m). This demonstrates the doping process of the chain by oxidation and the emergence of polaronic and bipolaronic bands, as discussed in Figs. 2 and 3.

4. Conclusion

The coexistence of phenylene and thiophene functionalities within the same polymer chain imparts a dual set of advantageous optical and electrical properties to the material. Consequently, the amalgamation of these attributes within a single polymer holds substantial promise for deployment in optoelectronic devices and as a pH-sensitive component. Additionally, it may serve a secondary, but significant, role as a sensor for ionizing gamma radiation. In this study, we have expounded upon the chemical synthesis of a novel polymer, PTBT46, characterized by the covalent integration of thiophene and phenylene units. The comprehensive physicochemical analysis conducted herein has unveiled intriguing phenomena. Notably, the pronounced alteration in the solution's color as a function of pH variation was observed. Furthermore, the in-depth investigation encompassed UV–Vis-NIR in situ measurements, applied under anodic potentials, which have authentically validated the phenomenon of chain doping within PTBT46. This validation is manifested through the appearance of discernible polaronic and bipolaronic bands in the material's optical spectrum, indicating unequivocally that these bands stem from oxidative processes affecting the polymer's molecular structure.

Further exploration of PTBT46′s properties in solution, utilizing both THF and Chloroform as solvents, uncovered an intriguing solvent dependence. This confirmation of solvent influence was established through absorbance measurements, which distinctly revealed the emergence of two new bands, centered at 680 nm and 2630 nm, within the chloroform solution exposed to doses ranging from 25 Gy to 100 Gy. Strikingly, these bands were conspicuously absent in the spectra of the THF solution. The appearance of these near-infrared bands in the absorbance spectra of polymers is commonly attributed to the presence of polarons and bipolarons, phenomena intimately linked with doping processes. Consequently, the color shift observed in the PTBT46/$CHCl_3$ solution post-irradiation can be conclusively interpreted as a manifestation of the doping effect. This phenomenon resonates with prior knowledge derived from electrochemical studies of conjugated polymers, notably within the context of electrochromism.

In conclusion, compounds based on thiophene-phenylene groups in halogenated solutions are readily susceptible to doping through gamma radiation. Doping processes in polymers, whether achieved chemically or electrochemically, may introduce non-uniform dopant distributions, with a challenging control in the chemical case and often leading to limited doping in the electrochemical scenario. Conversely, gamma radiation-induced doping, administered at controlled doses, establishes a unique correspondence between radiation exposure and doping level. This method enables facile doping of large quantities, though it's essential to acknowledge that not all polymers are amenable to such doping, as the process is reliant on specific chemical mechanisms inherent to each material.

The study conducted herein demonstrates that polymers based on thiophene-phenylene exhibit a distinctive ionizing radiation-induced doping behavior when contrasted with the behavior of MEH-PPV, wherein polymer chain scission occurs. In PTBT46, the observed effect is a reduction in the degree of conjugation owing to conformational alterations. The amalgamation of thiophene and phenylene constituents within the same polymer chain profoundly influences the doping process, influenced by the presence of thiophene groups and methoxy moieties, as well as S…O interactions.

Anions manifest a predilection for attacking the methyl group, directly interfering with S…O interactions, potentially disrupting them and thereby inducing conformational defects. Concurrently, this results in chain doping via alteration of the chemical environment proximal to the thiophene group, culminating in the formation of polarons and bipolarons. Consequently, significant changes in optical spectra were observed, accompanied by a substantial increase in PTBT46′s conductivity with increasing gamma radiation dosage. FTIR measurements further corroborate the hypothesis of effective conjugation alteration in PTBT46 due to conformational changes, with notable shifts in intensity in specific infrared absorption bands. These changes align with the model proposed in this study, wherein radicals stemming from chloroform radiolysis primarily attack the methyl group, disrupting S…O interactions and elucidating alterations in relative intensity in the asymmetric stretching of C–C bonds between rings at 1272 cm^{-1}. This alteration is closely linked to conformational defects rather than polymer chain breakage, as observed in the case of MEH-PPV.

In summary, the interaction between the polymer and gamma radiation follows a sequential chain of events: initially, the radiation breaks down the chloroform molecules, generating radicals (Cl• and $CHCl_2$•) in the solution. These radicals then attack the polymer chain, initiating an oxidation process that leads to the formation of polarons and bipolarons. These species serve as dopants for the polymer, increasing its conductivity and inducing distortions in the network structure, consequently reducing the effective conjugation length within the polymer. The gamma radiation-induced breakdown of chloroform also produces HCl in a subsequent step. pH measurements revealed an increase in the acidity of the polymer solution as the radiation dose increased, confirming the chloroform breakdown and the presence of radicals as dopant agents. The same effect was also chemically achieved by doping the PTBT46/THF solution with HCl. In this experimental study, materials based on thiophene and phenylene have been demonstrated as a novel class of semiconducting polymers. In this innovative approach, gamma radiation serves as a tool for the doping process, leading to a significant enhancement in conductivity. This opens up new avenues for conducting polymers, including applications in optoelectronics and radiation sensing.

CRediT authorship contribution statement

M.D.R. Silva: Writing – original draft, Methodology, Investigation, Conceptualization. **Silésia de Fátima Curcino da Silva:** Writing – review & editing, Writing – original draft, Visualization, Formal analysis, Data curation, Conceptualization. **T. Jarrosson:** Resources. **J.-P. Lère-Porte:** Resources. **Françoise Serein-Spirau:** Resources. **Roberto S. Nobuyasu:** . **Gustavo G. Dalkiranis:** . **Bruno S. Zanatta:** . **Sandra Lucia Nogueira:** Writing – review & editing, Writing – original draft, Formal analysis. **M. Foschini:** Writing – original draft. **R.A. Silva:** Resources, Funding acquisition. **A. Marletta:** Writing – original draft, Validation, Supervision, Resources, Project administration, Funding acquisition, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.eurpolymj.2024.113485.

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